NARRATIVE LITERATURE REVIEW

# Published Models of Weather, Thermal Strain and Human Health

Physiological mechanisms, cardiovascular and respiratory outcomes, and weather-associated symptoms

**Abstract.** Meteorological conditions enter several distinct classes of human-health models. This narrative review examines published thermoregulation models, physiological strain and comfort indices, environmental time-series methods, and selected applications to cardiovascular, respiratory, neurological and renal outcomes. The purpose is to identify genuine mathematical sources rather than construct a new predictive system. Sources comprise original methodological papers, empirical studies, associated public implementations and a separately identified international standard. No participant data were collected, no coefficients were estimated and no meta-analysis was performed. Fiala, JOS-3 and predicted heat strain address physiological responses; distributed lag nonlinear models and case-crossover analyses address outcome-specific exposure associations. The evidence does not establish one transportable equation for every disease or a universally valid personal thermal-endurance countdown. Published equations are re-typeset with precise source locations. Limitations in individual prediction, access to fitted parameters and internal reporting are distinguished from the existence of peer-reviewed publication.

**Keywords:** human biometeorology; thermal tolerance; predicted heat strain; cardiovascular disease; asthma; COPD; distributed lag nonlinear model; weather-associated pain.

## 1 Introduction

A weather forecast and a health forecast answer different questions. The first describes an atmospheric state; the second relates exposure to a specified physiological response, symptom, admission or death. The literature therefore contains several mathematical objects rather than one interchangeable “weather-health formula.” Thermoregulation models describe bodily heat exchange and control, thermal indices summarize a specified environment or response, and epidemiological models estimate associations from observed outcomes. Their outputs and validation criteria differ [1, 2, 5, 10].

The word *prediction* also has several meanings. A model may reproduce a laboratory temperature trajectory, explain historical changes in hospital counts, estimate a relative risk at a selected exposure, or forecast a new individual’s event. None of these achievements automatically demonstrates the others. This distinction is essential when assessing disease-history-based advice: a thermal index is not a diagnosis, and a city-level mortality curve is not an individual probability calculator [3, 14, 24].

## 2 Approach and interpretation of evidence

Targeted web discovery, exact-title searches and citation follow-up were used to locate publisher pages, PubMed-indexed records, institutional or author manuscripts and original

software. Sources were checked on 13 October 2026. Some articles were available in full, whereas others could only be verified through an abstract, institutional record or associated code. Those distinctions are retained in the companion evidence review. A public landing page is not a promise that its PDF can always be downloaded.

Coverage is selective rather than exhaustive; patents and newly invented formulas are not treated as peer-reviewed model evidence. This is a narrative synthesis, not a prospectively registered systematic review [25]. There was no independent dual screening, formal risk-of-bias scoring or pooled effect estimation. No new experiment, patient-level analysis or operational forecast was undertaken. “Published” identifies the source; “validated” is used only in relation to the outcome and test actually described. The reporting principles in TRIPOD+AI help distinguish clinical prediction development and evaluation from broader statistical associations [24].

All displayed equations below are existing published or publicly implemented formulas. Their numbering in this document is editorial. A source locator is supplied under every equation. When a complete equation system could not be verified, the model is described and referenced rather than reconstructed.

## 3 Thermal physiology, exposure duration and discomfort

### 3.1 Fiala, UTCI and physiological validation

The UTCI-Fiala model represents internal heat transfer, environmental heat exchange and thermoregulatory responses across a multi-node body representation. The UTCI operational procedure converts the modeled response into an equivalent thermal environment. Temperature, humidity, wind and radiation therefore belong to the thermal problem; “thermal” does not mean air temperature alone [1, 2]. UTCI should not be interpreted as an equation for myocardial infarction, asthma or a personal survival duration.

Psikuta et al. provide a separate validation study of the Fiala model using human experimental data. The reported overall root-mean-square deviations were $1.35 \pm 1.00\,^{\circ}\mathrm{C}$ for mean skin temperature and $0.32 \pm 0.20\,^{\circ}\mathrm{C}$ for core temperature. These are measures of physiological agreement under the studied conditions, not sensitivity or calibration for clinical events [3].

### 3.2 JOS-3 and reproducible thermoregulation

Takahashi et al. describe JOS-3, an 83-node thermoregulation model with publicly released code. The published implementation includes personal and environmental inputs and predicts physiological temperatures and regulatory responses [4]. The code makes algorithm inspection possible, an important advantage for a literature-to-reproduction workflow. It does not, by itself, establish validated disease-specific predictions for people with cardiopulmonary illness.

### 3.3 Predicted heat strain and the endurance-time question

Malchaire et al. developed and tested the predicted heat strain (PHS) model using 909 laboratory and field experiments. The model calculates time-dependent sweat loss and rectal temperature, supporting assessment of exposure duration under specified conditions [5]. ISO 7933:2023 is the contemporary standard reference for PHS calculations, but it is a stan-

dard rather than a peer-reviewed journal article, and its full computational text was not reproduced here [6].

PHS is the closest published family in this review to the question, "How long can a person remain in a hot environment before specified strain criteria are reached?" Its limiting duration is conditional on model inputs and criteria. It is not a personal survival clock and should not be relabeled as the maximum safe time for every individual with a disease. The 2001 validation paper and the 2023 standard are different documents; agreement with one should not be assumed to verify every later implementation [5, 6].

### 3.4 A genuine physiological strain equation

The physiological strain index (PSI) combines measured rectal temperature and heart rate [7]. It is a monitoring index, not a weather-only forecast:

$$\mathrm{PSI} = 5\frac{T_{\mathrm{re},t} - T_{\mathrm{re},0}}{39.5 - T_{\mathrm{re},0}} + 5\frac{\mathrm{HR}_t - \mathrm{HR}_0}{180 - \mathrm{HR}_0}. \tag{1}$$

*Source:* Moran et al. [7], original Eq. (4); also explicitly reproduced under "Calculations / Physiological strain index" in Davey et al. [8]. Fractions replace algebraically identical inverse factors; the model is unchanged.

Here $T_{\mathrm{re},t}$ is rectal temperature at time $t$ in degrees Celsius and $\mathrm{HR}_t$ is heart rate in beats per minute; the subscript $0$ denotes baseline. The constants 39.5 and 180 are normalization constants in the published index, not universal safe limits for individual users. A later experimental investigation found that absolute PSI did not reliably identify individuals reaching a thermal tolerance limit [8]. Publication of a compact equation is therefore not sufficient evidence for a personal stopping rule.

### 3.5 Thermal discomfort is a separate endpoint

The predicted percentage dissatisfied (PPD) relation is available in the public implementation associated with the peer-reviewed *pythermalcomfort* software paper [9]:

$$\mathrm{PPD} = 100 - 95\exp\left(-0.03353\,\mathrm{PMV}^4 - 0.2179\,\mathrm{PMV}^2\right). \tag{2}$$

*Source:* The public `pmv_ppd_iso` implementation associated with *pythermalcomfort* [9], specifically the assignment to `ppd_array`. This is an existing implemented relation, not a new fit or a numbered equation attributed to the software article.

PMV is the predicted mean vote used by the comfort model. PPD describes a modeled percentage dissatisfied with the thermal environment, not a percentage expected to develop disease. The formula is conditional on the PMV framework and applicable assumptions; it should not be applied as a general outdoor illness score. This transcription retains the numerical constants of the existing implementation.

# 4 Published mathematical foundations for weather–disease associations

## 4.1 The time-series mean model

Gasparrini, Armstrong and Kenward express the basic outcome model as follows [10]:

$$g(\mu_t) = \alpha + \sum_{j=1}^{J} s_j(x_{tj}; \boldsymbol{\beta}_j) + \sum_{k=1}^{K} \gamma_k u_{tk}. \tag{3}$$

*Source:* Gasparrini, Armstrong and Kenward [10], original Eq. (1), p. 2225.

The mean $\mu_t$ is $E(Y_t)$; $g$ is the link function; $s_j$ represents a modeled exposure response; and $u_{tk}$ denotes additional predictors. For daily health-event counts, their discussion uses an overdispersed Poisson setting and a log link. This is a published model family. It does not specify one universal set of disease coefficients.

## 4.2 The distributed lag nonlinear cross-basis

The same paper combines exposure and lag bases in the following expression:

$$s(x_t; \boldsymbol{\eta}) = \sum_{j=1}^{v_x} \sum_{k=1}^{v_\ell} \mathbf{r}_{tj.}^{\mathsf{T}} \mathbf{c}_{.k}\, \eta_{jk} = \mathbf{w}_{t.}^{\mathsf{T}} \boldsymbol{\eta}. \tag{4}$$

*Source:* Gasparrini, Armstrong and Kenward [10], original Eq. (7), p. 2227. Boldface and transpose symbols are typographical renderings of the original vector notation.

The vector $\mathbf{r}_{tj.}$ contains lagged, exposure-basis-transformed values, $\mathbf{c}_{.k}$ belongs to the lag basis, and $\eta$ contains estimated coefficients. The integers $v_x$ and $v_\ell$ give the dimensions of those bases. The final form is the cross-basis design vector multiplied by its coefficients. The `dlnm` paper and author-provided code supply practical implementation resources [11].

Lag length, basis type, knot placement and the reference exposure are parts of a fitted model, not natural constants. A lagged association is also not the same thing as a forecast lead time. For example, a study using the preceding 14 days of exposure has not thereby demonstrated accurate 14-day-ahead forecasts. The distinction follows from what is conditioned on and what is predicted.

## 4.3 Case-crossover and pooling in actual applications

Published disease applications also use within-person or time-stratified case-crossover comparisons and hierarchical pooling. Mukamal et al. compare transient exposures around severe-headache presentations; Dixon et al. compare pain-event and control periods; Alahmad et al. use city-level conditional count models and pooled exposure–response estimation [19, 20, 14]. These are documented applications, not newly suggested formulas. Their methods sections should be used for exact likelihoods and adjustment choices rather than importing an unrelated generic survival equation.

# 5 Cardiovascular outcomes

## 5.1 Myocardial infarction

Bhaskaran et al. analyzed 84,010 myocardial infarction admissions in 15 conurbations in England and Wales. The time-series regression considered temperature effects over the

current and following 28 days. Lower temperature was associated with greater myocardial infarction risk in that setting; the analysis did not identify a corresponding heat increase [13]. This is outcome-specific empirical evidence, not a universal rule that cold or heat affects all cardiovascular endpoints identically.

Its central modeling contribution is the use of defined clinical events and delayed temperature effects. A reported relative risk cannot be used alone to calculate a new person's absolute probability without the appropriate baseline risk and a validated prediction setting. No new infarction equation or fitted numerical coefficients are constructed here.

### 5.2 Cause-specific cardiovascular mortality

Alahmad et al. examined cardiovascular mortality across 567 cities in 27 countries. Their approach used city-specific temperature–lag relationships and hierarchical pooling; the main lag period was 14 days. Outcomes included ischemic heart disease, stroke, heart failure and arrhythmia, alongside total cardiovascular mortality [14]. The inclusion of an outcome in the analysis should not be mistaken for evidence of an identical or statistically significant response for every cause.

The study estimates cause-specific mortality relationships, not the onset of every cardiovascular condition. Its regional structure is particularly relevant to transportability: a local reference temperature and pooled association are not automatically the correct inputs for another city's population or one patient's medical history.

### 5.3 Stroke subtypes and broader temperature–mortality methods

The subsequent multicountry stroke analysis distinguishes ischemic and hemorrhagic stroke deaths. It uses conditional quasi-Poisson models, nonlinear distributed lags and pooling across locations, with cumulative effects considered over 14 days [15]. This distinction matters because combining subtypes can hide differences in the outcome being modeled.

For a broader methodological precedent, the 2015 multicountry mortality study used location-specific Poisson time-series models, a 21-day DLNM and multivariate meta-regression [12]. The contrast between 14- and 21-day choices demonstrates that no single lag specification should be copied indiscriminately between studies. Both studies concern observed mortality associations; neither provides an externally calibrated tomorrow-risk equation for every individual.

## 6 Respiratory disease models

### 6.1 Heat-related respiratory admissions

Anderson et al. studied approximately 12.5 million Medicare beneficiaries in 213 US counties during 1999–2008 and estimated national respiratory-hospitalization associations using Bayesian hierarchical models. The population was older adults, and the outcome was emergency respiratory hospitalization [16]. Such estimates are informative for this population and endpoint, but not interchangeable with personal asthma symptoms, all-age COPD events or other admission definitions.

### 6.2 An explicit adult-asthma model

Zhang et al. examined 15,678 adult asthma admissions in Shanghai during 2005–2012, linked to meteorological and pollution data. Their published basic model is [17]:

$$\begin{aligned}\mathrm{Log}[E(Y_t)] = \alpha + \gamma\,\mathrm{DMT}_{t,l} + s(\text{time}, \text{df}) + s(\text{humidity}, \text{df}) \\ + s(\text{rainfall}, \text{df}) + s(\text{wind speed}, \text{df}) \\ + \beta\,\text{Holiday} + \varepsilon\,\text{DOW} + Z_t.\end{aligned} \tag{5}$$

*Source:* Zhang et al. [17], "Data analysis," unnumbered basic-model equation, PDF p. 2. Line breaks only are editorial. The original pollutant shorthand $Z_t$ is retained.

$Y_t$ is the daily asthma-admission count. $\mathrm{DMT}_{t,l}$ represents the daily-mean-temperature cross-basis with lag $l$ and coefficient vector $\gamma$; $s$ denotes spline adjustment. Calendar effects and the pollution contribution $Z_t$ are also included. In the printed expression, $\varepsilon$ accompanies day of week: it should not be reinterpreted as an added random residual.

The study allows lags up to 30 days. Humidity, rainfall and wind are adjustments in this model; their presence does not make them independent forecasts of asthma or prove that one dominates temperature. The pollutant notation is abbreviated in the original equation, and a full fitted model still requires basis settings and coefficient estimates. The available study describes association and diagnostic checks rather than external validation of a personal forecasting application [17].

### 6.3 A COPD example with reporting limitations

Li et al. describe quasi-Poisson GAM/DLNM analyses of COPD admissions in Qingyang, with temperature, diurnal temperature range and humidity as the primary exposures in separate models. The first printed formula is retained below because it illustrates both a published specification and the need to inspect reporting quality [18]:

$$\begin{aligned}\log(Y_t) = \alpha + \beta\,\mathrm{TEM}_{t,l} + \mathrm{ns}(\mathrm{PM}_{10t}, \text{df}) \\ + \mathrm{ns}(\mathrm{SO}_{2t}, \text{df}) + \mathrm{ns}(\mathrm{NO}_{2t}, \text{df}) \\ + \mathrm{ns}(\mathrm{RH}_t, \text{df}) + \mathrm{ns}(p_t, \text{df}) \\ + \mathrm{ns}(\mathrm{Time}_t, \text{df}) + \text{Dow} + \text{Holiday}.\end{aligned} \tag{6}$$

*Source:* Li et al. [18], "Statistical models," first unnumbered formula. The printed left side $\log(Y_t)$ is preserved, not silently changed to $\log E(Y_t)$. See the reporting cautions accompanying this transcription.

The article defines $\mathrm{TEM}_{t,l}$ as the temperature cross-basis, ns as a natural cubic spline, and $p_t$ as atmospheric pressure. There are two important reporting issues. The left side is printed as $\log(Y_t)$ although the surrounding text describes a quasi-Poisson count model. Also, pollutants are included in the formulas, while the limitations state that air-pollution confounding was not adjusted for. These inconsistencies should be resolved with the authors before exact reproduction; this review neither silently repairs the equations nor treats publication as verification of all details [18].

## 7 Headache, pain and other clinical outcomes

### 7.1 Severe headache and weather-associated pain

Mukamal et al. used a case-crossover design for 7,054 severe-headache emergency presentations, with temperature, pressure, humidity and air-pollution exposures considered

over short time windows. The endpoint was a presentation to emergency care, not every migraine episode experienced outside hospital [19]. A published correction accompanies this article; its existence is noted in the bibliography.

Dixon et al. analyzed daily pain reports from 2,658 participants over 15 months. Their case-crossover study examined humidity, wind speed, pressure and temperature, with modest weather–pain associations. The analysis does not demonstrate a separately calibrated forecast for each participant [20]. It provides a useful precedent for linking symptom reports with location-dependent weather, while keeping shared exposure associations distinct from personalized prediction.

### 7.2 Renal, fluid-balance and other admissions

Bobb et al. compared heat-wave and matched non–heat-wave periods across 1,943 US counties. They identified elevated admission risks for fluid/electrolyte disorders, renal failure, urinary tract infection, septicemia and heat stroke in older adults. They also reported lower congestive-heart-failure admission risk and did not reproduce an earlier respiratory association under their study definition [21].

This disagreement is informative, not a reason to average unlike estimates. Cause-specific death, hospitalization, continuous temperature and threshold-defined heat waves are different targets and exposure contrasts. A general claim that every heart or lung outcome increases in the same way would be inconsistent with the selected evidence. The scope of this review does not establish equations for every infection, pregnancy outcome, psychiatric condition or chronic disease.

## 8 Published personalized warning systems

ClimApp integrates individual factors and weather forecasts to provide thermal-stress warnings and guidance. HEAT-SHIELD describes occupational heat-risk forecasting with worker-related information and behavioral guidance [22, 23]. These are published precedents for personalized warning delivery rather than newly invented architectures in this review.

Their existence should not be equated with validated probabilities for all cardiovascular and respiratory events. An individualized thermal warning may account for body characteristics, clothing or activity while still lacking disease-specific clinical validation. The appropriate question is which outcome and population were evaluated, not whether the system uses the word “personalized.”

## 9 Discussion: what can be carried forward into research

### 9.1 There are several kinds of mathematical credibility

An inspected equation provides *specification*. Public code provides another layer of *reproducibility*. Agreement with physiological measurements provides *validation for those measurements*. Association with clinically coded counts provides *epidemiological evidence*. None alone guarantees an accurate, calibrated or beneficial personal warning. This distinction follows directly from the different outputs of the thermal, cardiovascular and symptom studies reviewed here [3, 8, 14, 20].

A literature review can therefore be substantive without experiments. It can establish what was modeled, identify how variables and lags were defined, locate equations and software, and critically distinguish reported testing from untested applications. It should not manufacture coefficients to make an article seem operationally complete.

### 9.2 No universal ranking of weather variables

The reviewed models treat meteorological variables differently. Temperature, radiation, air movement and humidity jointly enter thermal assessment. Temperature is the focal exposure in several cardiovascular analyses. Humidity, pressure and wind feature prominently in the pain study, while the asthma model adjusts for additional weather and pollution variables [2, 14, 17, 20]. This supports an outcome-specific literature organization rather than a fixed weighting such as "temperature accounts for half of health risk." No such universal weights are supplied here.

### 9.3 Boundaries for individual prediction and advice

Clinical prediction claims require an explicit outcome, horizon, intended population and evaluation procedure. TRIPOD+AI supplies reporting guidance for prediction studies, including regression and machine-learning methods; it is not proof that any model is clinically useful [24]. Similarly, advice effectiveness is not established by a statistically significant exposure coefficient. This review deliberately stops short of medical recommendations, invented individualized thresholds or a new hybrid risk equation.

## 10 Conclusions

The literature provides genuine mathematical sources for thermal strain, discomfort, delayed environmental associations and selected disease outcomes. PHS addresses limiting exposure duration under specified assumptions; Fiala and JOS-3 simulate physiological responses; PSI and PPD summarize distinct strain or comfort endpoints; and DLNM, case-crossover and hierarchical count analyses quantify outcome-specific weather associations. The evidence reviewed does not establish a universal personal disease predictor or survival countdown. The defensible research contribution at this stage is a transparent comparison of published models, their exact specifications, supporting observations and limits of applicability.

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